

A Modeling Method for Conceptualizing and Documenting Software Architectures

Sven Christ

Preface

XXX

Contents

1	Introduction	1
	Bibliography	2

Preliminary Philosophical Considerations

A peculiar intellectual roadblock halted my initial attempt to conduct this research even before it began. The fundamental question I faced concerned the nature of research itself. How could I proceed to conduct research without first explicating what research precisely is? Upon inquiry, I came to regard research as the process of attaining the aim of any scientific endeavor: *expanding the body of human knowledge*.

This definition, however, generated a vexing new question and led me down the rabbit hole of epistemology: *What is knowledge?* Linguistically, this question constitutes a sensical inquiry. To answer it — or any question — I would need to *know* the answer, thus employing knowledge to define the concept of knowledge, which is evidently nonsensical.

The reader might possess a peripheral awareness of the problem I faced and perhaps has conquered this obstacle in their own work in some way or another. I, however, could not proceed without addressing this foundational issue.

Motivation

This dissertation is a scientific endeavor. The aim of any scientific endeavor is the expansion of knowledge. To expand knowledge, we must first define what knowledge is.

That is, what can and what can not be known.

The expansion of knowledge – the aim of any scientific endeavor – requires a prior definition of the concept of knowledge.

How, though, can we know what knowledge is?

To pose a research question, we must first distinguish the space of sensical answers from the non-sensical and senseless answers. To configure and execute a research method, we must first

To contribute to the existing body of knowledge.

I will not explicitly delve into the different epistemological positions. And neither give an overview because it is obsolete. I will argue my case through which my positioning will become apparent.

I confess, I can – in the format of this text – use mere words to communicate my thoughts. My first request, thus, is for you to conspire with me and accept that what I will attempt to say, can, in fact, be verbalized. How, otherwise, may the nature of knowledge be examined when the words employed to that end are required to be true and thus contain knowledge themselves?

Note: I consciously decided on the term *word*, to differentiate it from *language* and other symbols. See, also, the “Triangle of Reference” (Ogden et al. 1923).

The Postulate of Shared Meaning

To begin, let us face a meta-ontological question and ask how we can define (in words) any concept (including knowledge) that we can grasp with our mind. Axiomatically, we must accept that our cognition will map the concepts in our mind *appropriately* onto the words we say. This statement remains a necessary assumption for any well-meaning discussion. Any attempt to investigate what constitutes this appropriateness (i. e. circumstances, time, individualities) is futile, because the spoken words are the only gateway to the mind of the speaker (Wittgenstein 1953). Any attempt to skip the words and move our focus towards our own mind would inevitably lead to methodological solipsism (Putnam 1975, p. 136), which would render any communication about the concepts contained within the mind redundant.

XXX TODO: Show that evaluating pure truth of linguistic statements is impossible because the definitions of words that are used to express the statements that needs to be evaluated need to be given ad infinitum XXX

It follows, then, that words can only obtain *meaning* when they – not their truth, but rather their sense – can be validated against something outside the mind of the speaker. That is, when no one hears me, speaking itself becomes nonsensical (Wittgenstein et al. 1933). I can call the sky “blue” today, and “green” tomorrow and “red” the day after. And each statement would be true and would have to pass as knowledge, because there is no way (and no need) to proof it. Everything I would say under these circumstances becomes true by default.

We have arrived at my first epistemic show-of-hands: The postulate of *shared meaning*, which states that knowledge can only exist intersubjectively. Which means, that the words I use to verbalize my mind’s concepts must correspond – when listened to – to *equivalent* concepts in the mind of the listener. How can we be sure of that?

The Postulate of Atomicity

XXX Maybe instead of Atomicity: The Postulate of Pragmatism - knowledge must lead to justified action XXX

Even with no other human listening to us, we can never speak into the void. Nature will always be there. We can not lie to nature nor escape it. However, nature does not speak any natural language. Thus, we can not ask it to explain itself. We may only ever ask: yes or no? And we do that by executing actions against nature and investigating the resonance. Over time we can build theories about nature and express them in words under the assumption that other humans interpreting these words have the same understanding and use the same words to describe the phenomena witnessed in nature (Aristotle 1933, 1011b). This is the basis for any research in the natural sciences. Because the static and tangible subject of research allows us to build theories and retreat to the methods of fallibilism. The natural sciences escape any definition of knowledge by this. Instead, building their argument on top of pragmatism. Can we use the same methodological approach and the same epistemological paradigms when considering a contingent research subject like the human mind (Frank 2007)? See, “Correspondence Theory of Truth” (David 2022).

So, again, how can we be sure that the words we speak will evoke the same mental concepts in the listener as in ourselves? To repel any ad absurdum discussions about the “nature of reality”, let us also please follow a phenomenological approach to this question (Husserl 2002). See, “Twin Earth thought experiment” (Putnam 1973).

One approach is to, again, consider words to be our portal into the human mind (see above) and enter a *dialogue* with other humans about what constitutes knowledge. But which words – which knowledge – will ultimately be accepted? We will accept that, which is most useful and effective to facilitate successful actions in practical contexts. This means, that truth is not an abstract, static concept but is determined by the practical outcomes of holding certain beliefs. See, the “Pragmatic Theory of Truth” (Capps 2023).

This leads to my second epistemic unveiling: The postulate of *atomicity*, which states that knowledge exists *sui generis*. Knowledge is not contingent upon external validation or empirical evidence, but on its inherent capacity to produce practical outcomes. We may not break knowledge down into further components like justification and belief. See, “Knowledge and its Limits” (Williamson 2002).

But what is a “problem” and a “successful action”? Epistemology stops here and answers with *belief*. We would have to enter the realm of existentialism to find answers, which I will do – briefly. The question is, how can we construct a bridge to close the gap between the pragmatism required for knowledge and the evaluation of proof of

pragmatic ideas when we have no mirror – like nature – against which to check our knowledge.

The Postulate of the Will

When knowledge can only be expressed in words, which it has to, as we have seen above. Only under the assumption that a limited number of word configurations exists, can we speak of “contributions to the body of knowledge”. It appears, though, that this assumption does not hold. First, even the countable set of words, that we call our vocabulary, grows and I can see no inherent growth throttling of this set - a priori or empirically. Second, the set of combinations of words – their configuration – is infinite. Quod erat demonstrandum.

We could only bypass this by accepting that our mind itself is limited in its capacity to create pictures out of these word configurations and that eventually clusters of pictures emerge that are in themselves “similar enough” to speak of equivalency between them, inside the clusters. This does however require a similarity metric - not necessarily quantifiably stated. How? With the aim of advancing one’s will (Kant, Schopenhauer, Nietzsche, Sartre). In this sense, as the will is continuously evolving and at any point there exists one thinkable yet unattainable optimal solution to advancing it, knowledge is in any point limited but required to be extended when considered on a continuum.

This leads me to my final epistemic unraveling and my confession to following a perspectivistic approach to anything that comes after this: The postulate of *the will*, which states that it is our individual will to interpret and give meaning to our experiences and actions (Schopenhauer 2009).

To conclude, I can only try to communicate my own *perspective* and try to argue, to the best of my ability, how pragmatically useful my thoughts are – for me. I can not – nor will I try – to convince the reader of anything that is outside the scope of my perspective.

XXX TODO: Try to differentiate or integrate *usefulness* in i. e. information systems
XXX

Summary

I wanted to define what knowledge is. I have first followed a philosophy of language approach and asked how we can define, using words, *anything*. We have seen that we need shared meaning is a requirement.

How can we proof that we actually mean the same thing by the words we say? I have then borrowed methods from the natural sciences and stated that it is the pragmatic content that makes knowledge knowledge. This means, knowledge is not something that statements possess but rather that knowledge is the thing that has the most benefit for our actions. This also means that we can never know anything for certain. We can just know *enough* for a specific *goal*.

To advance these thoughts one step further and answer what pragmatism is and by what metric it can be evaluated, I have refuted blind faith and looked into the philosophy of existentialism and employed the concept of the *will* as propagated by prominent philosophers. Will is something highly individualistic that can not nor should it be further investigated.

In conclusion, I must follow a perspectivistic approach in this dissertation and tell how things appear to *me* and what insights *I* need to advance *my* will. Fear not, though. As we are all humans and my readers will supposedly be scholars like myself, it is fair to assume that my insights will stem from an appropriately similar will and thus exhibit enough congruency for them to be useful enough for my readers to consume them.

XXX TODO: Show that the quest for originality or generalization is unnecessary: Even if the meaning of what I say has been said before, I have now used different words. That is original (enough). What scope of generalization are we talking about?
XXX

1 Introduction

Bibliography

- Aristotle (1933). *Aristotle in 23 Volumes*. Trans. by H. Tredennick. Vol. 17, 18. Cambridge, MA; London: Harvard University Press; William Heinemann Ltd.
- Capps, J. (2023). “The Pragmatic Theory of Truth”. In: *The Stanford Encyclopedia of Philosophy*. Ed. by E. N. Zalta and U. Nodelman. Summer 2023. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/sum2023/entries/truth-pragmatic/>.
- David, M. (2022). “The Correspondence Theory of Truth”. In: *The Stanford Encyclopedia of Philosophy*. Ed. by E. N. Zalta. Summer 2022. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/sum2022/entries/truth-correspondence/>.
- Frank, U. (2007). “Ein Vorschlag zur Konfiguration von Forschungsmethoden in der Wirtschaftsinformatik”. In: *Wissenschaftstheoretische Fundierung und wissenschaftliche Orientierung der Wirtschaftsinformatik*. Ed. by F. Lehner and S. Zelewski. Berlin: GITO, pp. 155–184.
- Husserl, E. (2002). *Ideen Zu Einer Reinen Phänomenologie Und Phänomenologischen Philosophie*. Berlin, Boston: De Gruyter. DOI: doi:10.1515/9783110916096.
- Ogden, C. K. and I. A. Richards (1923). *The Meaning of Meaning: A Study of the Influence of Language upon Thought and of the Science of Symbolism*. 1st ed. New York: Harcourt, Brace & World, Inc.
- Putnam, H. (1973). “Meaning and Reference”. In: *The Journal of Philosophy* 70.19, p. 699. DOI: 10.2307/2025079.
- Putnam, H. (1975). “The Meaning of ”Meaning””. In: *Language, Mind, and Knowledge*. Minnesota Studies in the Philosophy of Science 7. Minneapolis: University of Minnesota Press, pp. 131–193.
- Schopenhauer, A. (2009). *Die Welt als Wille und Vorstellung*. Köln, Germany: Anaconda Verlag.
- Williamson, T. (2002). *Knowledge and Its Limits*. Oxford University Press. DOI: 10.1093/019925656X.001.0001.
- Wittgenstein, L. (1953). *Philosophical Investigations*. Ed. by G. E. M. Anscombe. New York, NY, USA: Wiley-Blackwell.
- Wittgenstein, L., G. C. M. Colombo, and B. Russell (1933). *Tractatus Logico-Philosophicus*. Ed. by C. K. Ogden. New York, Harcourt, Brace.